

Question Paper

Exam Date & Time: 23-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Communication Theory [ECM_3121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain what is a random process and wide-sense stationarity (WSS) random process. Explain why the assumption of WSS is useful in real-life communication and signal-processing applications. (4)

A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) A continuous random variable X has the probability-density function (4)

$$f_X(x) = ae^{-b|x|}, -\infty < x < \infty, \text{ where } a \text{ and } b \text{ are constants.}$$

Apply the normalization condition for a valid PDF and determine the relationship between a and b , given that $a, b > 0$.

Using the definition of cumulative distribution function

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \text{ derive the CDF and show all intermediate steps clearly.}$$

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) Explain the concept of uniform distribution and state its mathematical description with relevant parameters. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Illustrate the major blocks of a digital communication system with a neat labelled diagram. Explain the function of each block. (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Consider an interval Δt of FM wave $s(t) = A_c \cos(\theta(t))$ such (4)

that $\theta(t)$ in radians, satisfies $\theta(t + \Delta t) - \theta(t) = \pi$. Show that if Δt is sufficiently small, the instantaneous frequency of the FM wave is approximately given by $f_i \approx 1/(2\Delta t)$

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) Explain the concept of amplitude modulation (AM) and derive its mathematical expression. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Illustrate and explain the sampling theorem for an arbitrary signal $x(t)$. Analyze what happens when the criterion is not satisfied. (4)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Given $x(t) = \cos(150\pi t) + \cos(700\pi t)$. a) Determine the highest frequency content. b) Calculate the Nyquist rate. c) If sampled at $f_s = 500$ Hz, identify and explain the observed frequencies with appropriate illustrations. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Represent bitstream 01001011 by NRZ-M, Bipolar RZ, and Manchester line coding schemes. Describe the requirements for each scheme in a line or two. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Explain the concept of Entropy. State its mathematical description and any two properties. (2)

A)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) Consider the discrete memoryless source with source alphabet $A = \{s_0, s_1, s_2\}$ with their corresponding probabilities $p_0, p_1, p_2 = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$. Calculate the entropy. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Compute the Huffman code for symbol set with probabilities $\{s_0, s_1, s_2, s_3, s_4\} = \{0.3, 0.3, 0.2, 0.1, 0.1\}$. Compute the average codeword length and entropy. In the combined symbol, make moves as high as possible. Repeat by moving as low as possible. Compute variance for both approaches and compare (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) I) Explain the concept of generator matrix and parity check matrix. State their significance in a transmit and receive subsystem. (5)

A)

II) List any 3 types of cyclic codes.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) For a (7,4) cyclic code, the polynomial $(1+D^7)$ is factorized as $(1+D)(1+D^2+D^3)(1+D+D^3)$. Determine the parity check polynomial for the generator polynomial $(1+D+D^3)$. Compute the message and codeword polynomial, for the message sequence (1100). (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECM_3121:Communication Theory

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10

3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECM_3121:Communication Theory

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	6	18(36%)
4	Analysis	7	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	50 (100%)